

The equation $y(x^3 - z^2) = 2x - 1$

An exact reduction and verified partial results

Status. This note does *not* establish the requested infinitude. It gives self-contained elementary results, an exhaustive computation in a stated range, and an exact obstruction to a promising construction. Neither infinitude nor global finiteness is proved here. No internet searches were used.

1. The decisive divisibility condition

Write

$$d = x^3 - z^2.$$

Since $2x - 1 \neq 0$ for integral x , both y and d are nonzero, and

$$d \mid 2x - 1, \quad z^2 = x^3 - d, \quad y = \frac{2x - 1}{d}. \quad (1)$$

Conversely, every integer x and nonzero signed divisor d satisfying (1) with a square $x^3 - d$ give a solution. Thus this is an exact equivalence, not just a necessary test.

For $x \geq 1$, it implies

$$0 < |x^3 - z^2| \leq 2x - 1, \quad x^3 - 2x + 1 \leq z^2 \leq x^3 + 2x - 1. \quad (2)$$

The intuition is therefore to construct a cube unusually close to a square and simultaneously make their difference divide $2x - 1$. Closeness alone does not ensure the divisibility.

Proposition 1. *In every solution, y and d are odd, x and z have opposite parity, and $\gcd(x, z) = 1$. Moreover,*

$$(4z)^2 \equiv 2 \pmod{|d|}.$$

Proof. The product $yd = 2x - 1$ is odd. Reducing $d = x^3 - z^2$ modulo 2 gives the parity assertion. If a prime divided both x and z , it would divide d and $yd = 2x - 1$, an impossibility. Finally, $2x \equiv 1 \pmod{|d|}$ and $z^2 \equiv x^3 \pmod{|d|}$ imply $8z^2 \equiv 1 \pmod{|d|}$; doubling gives the displayed congruence. $\square$

The following are the verified solutions in the computed range $1 \leq x < 3,000,000$, together with all solutions for $x \leq 0$:

x	y	z	$d = x^3 - z^2$	$yd = 2x - 1$
-1	3	0	-1	-3
0	1	± 1	-1	-1
1	1	0	1	1
2	-3	± 3	-1	3
32	9	± 181	7	63
43	-5	± 282	-17	85

The table lists ten triples. It is not asserted to be globally complete.

2. Three complete elementary classifications

Proposition 2. *The solutions with $x \leq 0$ are exactly*

$$(-1, 3, 0), \quad (0, 1, \pm 1).$$

The only solution with positive square x is $(1, 1, 0)$. The solutions with $y = 1$ are exactly $(0, 1, \pm 1)$ and $(1, 1, 0)$.

Proof. *Nonpositive x .* If $x = -a$ with $a \geq 1$, the equation becomes

$$y(a^3 + z^2) = 2a + 1.$$

Thus $y \geq 1$ and $a^3 \leq 2a + 1$. For $a \geq 2$, $a^3 \geq 4a > 2a + 1$, so $a = 1$. Now $y(1 + z^2) = 3$ forces $1 + z^2 \in \{1, 3\}$; the latter would give $z^2 = 2$. Consequently $(x, y, z) = (-1, 3, 0)$. For $x = 0$, the equation is $yz^2 = 1$, giving $(0, 1, \pm 1)$.

Positive square x . Put $x = a^2$ with $a \geq 2$. The square a^6 has neighboring squares $(a^3 - 1)^2$ and $(a^3 + 1)^2$. Hence any nonzero difference $a^6 - z^2$ satisfies

$$|a^6 - z^2| \geq 2a^3 - 1 > 2a^2 - 1,$$

contradicting (2). For $x = 1$, the equation $y(1 - z^2) = 1$ gives either $y = 1, z^2 = 0$ or $y = -1, z^2 = 2$; only $(1, 1, 0)$ is integral.

The slice $y = 1$. The preceding arguments cover $x \leq 1$. For $x \geq 2$ we have

$$z^2 = x^3 - 2x + 1 = (x - 1)(x^2 + x - 1).$$

The two positive factors are coprime, because the second is congruent to 1 modulo $x - 1$. In a square, every prime exponent is even; since the factors are coprime, each factor must itself be a square. But

$$x^2 < x^2 + x - 1 < (x + 1)^2 \quad (x \geq 2),$$

which is impossible. □

3. Why a natural Pell construction fails

An attractive way to make a cube close to a square starts from the polynomial identity

$$(t^2 + 10t + 5)^3 - (t^2 + 4t - 1)^2(t^2 + 22t + 125) = 1728t. \quad (3)$$

This identity is verified by expanding both sides. If

$$s^2 - 10(t + 11)^2 = 40,$$

then the assignments

$$x = \frac{t^2 + 10t + 5}{10}, \quad y = \frac{25(t + 10)}{216}, \quad z = \frac{(t^2 + 4t - 1)s}{100} \quad (4)$$

satisfy the original equation over the rationals whenever t, s are rational. Indeed, (3) gives

$$x^3 - z^2 = \frac{216t}{125}, \quad y(x^3 - z^2) = \frac{t(t + 10)}{5} = 2x - 1.$$

The scaling is chosen so that $2x - 1$ contains the same factor t as the cube-square difference. This explains the appeal of the construction.

However, it gives *no integer triples*. If $y = k \in \mathbb{Z}$, then $t = 216k/25 - 10$, and substitution into (4) yields

$$x = \frac{46656k^2 - 54000k + 3125}{6250}.$$

The numerator is odd and the denominator is even, so x cannot be an integer. This obstruction holds even when rational parameters are allowed. Solving the auxiliary Pell equation therefore cannot repair this particular family. It does not exclude other constructions.

4. An exhaustive finite computation

For completeness, the following Python code reproduces the positive- x part of the table. It uses integer arithmetic throughout; `isqrt(n)` is the greatest integer whose square is at most n .

```
from math import isqrt

N = 3_000_000
solutions = []
for x in range(1, N):
    lower = x*x*x - 2*x + 1
    upper = x*x*x + 2*x - 1
    lo = isqrt(lower)
    if lo*lo < lower:
        lo += 1
    hi = isqrt(upper)
    for z in range(lo, hi + 1):
        d = x*x*x - z*z
        if d != 0 and (2*x - 1) % d == 0:
            solutions.append((x, (2*x - 1)//d, z))
print(solutions)
```

Its executed output is

```
[(1, 1, 0), (2, -3, 3), (32, 9, 181), (43, -5, 282)]
```

Both signs of every nonzero z give solutions. The endpoints of the loop over z are exactly the nonnegative integers allowed by (2). Every actual solution with $1 \leq x < N$ must therefore be tested, and the divisibility check is sufficient by (1). This proves the completeness of the search procedure within its stated range.

What remains unresolved in this answer. An infinitude proof must still construct infinitely many distinct integers x and nonzero divisors $d \mid 2x - 1$ for which $x^3 - d$ is a square, or supply an equivalent argument. The finite computation, the complete special slices, and the failed Pell construction supply no such proof. They also do not prove that only finitely many solutions exist.